

Roel Heremans

Senior data scientist

Senior data scientist with >10 years of industrial experience, in Artificial Intelligence and machine learning, with a background in High Energy Physics. Writing production- quality analytics code in various domains including Industry 4.0, Smart and Safe cities and Sports & Entertainment.

roel.heremans@gmail.com
+351 913 0000 48
Ponta do Sol, Madeira Island Portugal
linkedin.com/in/roel-heremans
www.roel-heremans.com

WORK EXPERIENCE

Freelance Data Scientist

01/2022 – current *Madeira, Portugal*

Key Projects

- **Bef4U Project:** Developed an XGBoost-based anomaly detection model for energy consumption in shopping malls.
- **Electrolux Project:** Built a Markov Random Fields model for service call forecasting, achieving 4.2% MAPE in predictive maintenance.
- **Magenta Insight LTD:** Designed an AI-driven system for detecting well water pumping via temperature sensors, with a Python-based dashboard.
- **Your Political Monitor:** Created an AI-powered web app for summarizing political messages using LangChain's Map-Reduce and Refine techniques.
- **The sniffers:** Processing Piping and Instrumentation Diagrams (P&ID) from autocad, to detect the symbols, and texts from the pdfs. Retraining a deep learning ResNet-50 model for object detection and a TrOCR model from Huggingface Transformers for the text recognition.
- **RoboVision:** Calculation of the VolumeFlow of material passing by a stereo camera setup, using classical opencv techniques to determine height profiles and deep learning approaches (CrocoFlow and MemFlow) for flow estimation.
- **LexGen:** Building evaluation framework for LLM in the field of legal questions. LangGraph was used for the Agentic workflow setup and LangFuse was used to visualize the model evaluation over time (after model updates).

Senior Data Scientist

AGT International

01/2015 – 12/2021 *Darmstadt, Germany*

Achievements/Tasks

- Led the development of the world's first AI-powered sensor-based bull scoring system for the Professional Bull Riders (PBR) organization
- Implemented the analytics solution for bull scoring, including sensor signal processing, feature extraction, and predictive modeling of ride difficulty.
- Delivered the project to customer satisfaction.
- Contributed to pre-sales activities, proof-of-concepts, and scope definition for projects in sports analytics (basketball, horse riding, surfing).

Contact: Christian Debes – Lead data scientist - (linkedin.com/in/christiandebes)

Senior Researcher

AGT International

01/2012 – 12/2014 *Darmstadt, Germany*

Achievements/Tasks

- Developed predictive maintenance models for heavy machinery and machine state estimation for German industrial clients.
- Built analytics solutions for a dike stability monitoring system, involving failure prediction, trend analysis, and environmental system simulation.

Contact: Joachim Schaper – Senior VP R&D - (linkedin.com/in/joachim-schaper-2212991)

Senior Researcher

Royal Military School

01/2003 – 12/2011 *Brussels, Belgium*

Achievements/Tasks

- Developed analytics code for various projects including automatic flood detection using radar imagery, non-destructive testing of material defects using Terahertz radiation, mine detection on the sea floor using synthetic aperture sonar imagery.

Contact: Prof. Alain Muls – Head CISS - (linkedin.com/in/alain-muls-32b312135)

EDUCATION

PhD in Elementary Particle Physics | High Energy Physics

Vrije Universiteit Brussel

SKILLS

Hard

- **Machine Learning & Deep Learning:** Signal & Image Processing, Predictive Maintenance, Anomaly Detection, Neural Networks, Hyperparameter Optimization, Forecasting
- **Natural Language Processing:** Text Classification, Sentiment Analysis, Text Generation, Language Modeling, Machine Translation, Transformers (BERT, GPT, T5, ViT, BART), Large Language Models
- **NLP Frameworks & Tools:** LangGraph, LangChain, LangFuse, RAG (Retrieval-Augmented Generation), Agentic Workflow, Hugging Face
- **Algorithms:** Clustering, Regression, Classification, Time Series Analysis, Information Retrieval
- **Additional Skills:** Statistical Analytics, Speech Recognition, Data Visualization

Soft

- **Strong Determination:** to Deliver Projects on Time and with High Quality
- **Client-Focused Approach**, Striving for Satisfaction by Exceeding Expectations
- **Excellent Communication** and **Collaboration Skills**, Ensuring Clear and Efficient Interaction with Stakeholders
- **Problem-Solving** and **Adaptability** to Meet Changing Requirements and Challenges
- **Written and Oral Reporting, Presentation Skills**

TECHNICAL SKILLS

Analytics | Programming

- **Programming Languages**: Python (pandas, scikit, numpy, matplotlib, plotly, ML flow, nltk, pytorch), SQL, DataBricks, Matlab, C/C++, Bash, Perl, Latex
- **Machine Learning Frameworks**: PyTorch, scikit-learn, TensorFlow
- **Cloud & Data Tools:** Google Colab, Runpod, DataBricks, Google Cloud
- **NLP Libraries & Frameworks:** Hugging Face, LangGraph, LangChain, LangFuse, RAG (Retrieval-Augmented Generation), Agentic Workflow
- **Visualization & Deployment Tools:** Streamlit, Gradio, Plotly, label-studio (for annotation purposes)

Development

- **Continuous Integration & Development:** Atlassian Tools (Jira, Confluence, Bamboo, BitBucket)
- **Version control:** Git, GitHub, GitLab
- **Agile Methodology:** Agile Scrum, Kanban, Sprint Planning
- **Test Driven Development:** Unit testing

LANGUAGES

Dutch and French
Native or Bilingual Proficiency

English and German
Full Professional Proficiency

Portuguese
Learning